Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$Ge($ II)compounds are powerful reducing agents whereas $Pb(IV)$ compounds are strong oxidants. It is because

(A) Pb is more electropositive than Ge

(B) ionization potential of lead is less than that of Ge

(C) ionic radii of Pb^{2+} and Pb^{4+} are larger than those of Ge^{2+} and Ge^{4+}

(D) of more pronounced inert pair effect in lead than in Ge

Q.2

Often a ground glass stopper gets stuck in the neck of a glass bottle containing $NaOH$ solution. This is due to :

(A) The presence of dirt particles in between.

(B) The formation of solid silicate in between by the reaction of SiO_2 of glass with $NaOH$.

(C) The formation of Na_2CO_3 in between by the reaction of CO_2 of air and $NaOH$.

(D) Glass contains a boron compound which forms a precipitate with the $NaOH$ solution.

Q.3

$PbF_4, PbCl_4$ exist but $PbBr_4$ and PbI_4 do not exist
because of

(A) large size of Br^- and I^-

(B) strong oxidising character of Pb^{4+}

(C) strong reducing character of Pb^{4+}

(D) low electronegativity of Br^- and I^-.

Q.4

$$Pb + \text{dil. } HNO_3 \xrightarrow{\text{Warm}} P + Q\uparrow + H_2O$$

incorrect statement for Q is

(A) Paramagnetic colourless gas

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(B) It is oxidized to paramagnetic coloured gas by air

(C) It combines with $Fe_2(SO_4)_3$

(D) It is also obtained by disproportionation of HNO_2

Q.5

A metal, M forms chlorides in its $+2$ and $+4$ oxidation states. Which of the following statements about these chlorides is correct?

(A) MCl_2 is more ionic than MCl_4

(B) MCl_2 is more easily hydrolysed than MCl_4

(C) MCl_2 is more volatile than MCl_4

(D) MCl_2 is more soluble in anhydrous ethanol than MCl_4

Q.6

Lead pipes are not suitable for drinking water because

(A) lead forms basic lead carbonate

(B) lead reacts with water containing air to form $Pb(OH)_2$

(C) a layer of lead dioxide is deposited over pipes

(D) lead reacts with air to form litharge

Q.7

Which out of the following compounds does not exist?

(A) BF_3

(B) $TlCl_3$

(C) $TlCl_5$

(D) Both (b) and (c)

Q.8

A solid element (symbol Y) conducts electricity and forms two chlorides YCl_n (colourless volatile liquid) and YCl_{n-2} (a colourless solid). To which one of the following groups of the periodic table does Y belong?

(A) 13

(B) 14

(C) 15

(D) 16

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Q.9

Which of the following conceivable structures for CCl_4 will have a zero dipole moment?

(A) Square planar

(B) Square pyramid (carbon at apex)

(C) Irregular tetrahedron

(D) None of these

Q.10

Least thermally stable is —

(A) CCl_4

(B) $SiCl_4$

(C) $GeCl_4$

(D) $GeBr_4$

Q.11

To a piece of charcoal, sulphuric acid is added. Then:

(A) there is no reaction

(B) water gas is formed

(C) SO_2 and CO_2 are evolved

(D) CO and SO_2 are evolved

Q.12

The correct statement with respect to CO is

(A) it combines with H_2O to give carbonic acid

(B) it reacts with haemoglobin in RBC

(C) it is powerful oxidising agent

(D) it is used to prepare aerated drinks

Q.13

With respect to graphite and diamond, which of the statement (s) given below is (are) correct?

1 . Graphite has higher electrical conductivity than diamond

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2 . Graphite is harder than diamond

3 . Graphite has higher C - C bond order than diamond

4 . Graphite has higher thermal conductivity than diamond

(A) 1,2

(B) 1,2,3

(C) 1,3,4

(D) 2,3,4

Q.14

Which of the following is similar to graphite?

(A) B

(B) BN

(C) B_2H_6

(D) B_4C

Q.15

The correct order of increasing $C-O$ bond length of CO, CO_2 and CO_3^{2-} is:

(A) $CO_3^{2-} < CO_2 < CO$

(B) $CO_2 < CO_3^{2-} < CO$

(C) $CO < CO_3^{2-} < CO_2$

(D) $CO < CO_2 < CO_3^{2-}$

Q.16

Which of the following statement(s) is / are incorrect for CO_2?
(i) In laboratory CO_2 is prepared by the action of dilute HCl on calcium carbonate
(ii) Carbon dioxide is a poisonous gas
(iii) Increase in carbon dioxide content in atmosphere lead to increase in green house effect.
(iv) CO_2 as dry ice is used as a refrigerant for ice cream and frozen food.

(A) (i) and (ii)

(B) Only (ii)

(C) (i), (ii) and (iii)

(D) (ii) and (iii)

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Q.17

Certain organic compound on combustion produces three gaseous oxides A, B and C. A and C turned lime water milky, B turned anhydrous $CuSO_4$ blue and C turned $K_2Cr_2O_7$ solution green. The elements present in organic compounds are

(A) C, N, O

(B) C, H, S

(C) C, H only

(D) C, S only

Q.18

The ions present in Al_4C_3 , CaC_2 and Mg_2C_3 are respectively

(A) $C^{4-}, C_2^{2-}, C_3^{4-}$

(B) $C_2^{2-}, C^{4-}, C_3^{4-}$

(C) $C_3^{4-}, C_3^{2-}, C^{4-}$

(D) $C_3^{4-}, C^{4-}, C_2^{2-}$

Q.19

The catenation tendency of C, Si and Ge is in the order $Ge < Si < C$. The bond energies (in $kJmol^{-1}$) of $C-C, Si-Si$ and Ge $-$ Ge bonds
are respectively

(A) $348, 297, 260$

(B) $297, 348, 260$

(C) $348, 260, 297$

(D) $260, 297, 348$

Q.20

Which of the following statements is false?

(A) Water gas is a mixture of hydrogen and carbon monoxide

(B) Producer gas is a mixture of CO and nitrogen

(C) Water gas is a mixture of water vapour and hydrogen

(D) Natural gas consists of methane, ethane and gaseous hydrocarbons.

Answers & Solutions

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Q.1 Answer:

of more pronounced inert pair effect in lead than in Ge

Solution:

Ge(II) tends to acquire Ge (IV) state by loss of electrons. Hence it is reducing in nature. Pb(IV) tends to acquire Pb (II) O.S. by gain of electrons. Hence it is oxidising in nature. This is due to inert pair effect.

Q.2 Answer:

The formation of solid silicate in between by the reaction of SiO_2 of glass with $NaOH$.

Solution:

$$SiO_2 + 2NaOH \rightarrow \underset{\text{Sodium silicate}}{Na_2SiO_3} + H_2O$$

Q.3 Answer:

strong oxidising character of Pb^{4+}

Solution:

F and Cl are more oxidising in nature and can achieve Pb in (IV) O.S. but Br_2 and I_2 can not achieve Pb in (IV) O.S. secondly Pb^{4+} is strong in oxidising nature and in its presence, Br^- and I^- can not exist.

Q.4 Answer:

It combines with $Fe_2(SO_4)_3$

Solution:

Here only statement in option C is incorrect as it combines with $FeSO_4$ and not it combines with $Fe_2(SO_4)_3$

$Pb + \text{dil. } HNO_3 \xrightarrow{\text{Warm}} Pb(NO_3)_2 + NO\uparrow + H_2O$

$NO\uparrow \xrightarrow[\text{R.T.}]{O_2} NO_2\uparrow$

paramagnetic, coloured gas

$HNO_2 \rightarrow HNO_3 + NO\uparrow + H_2O$

Q.5 Answer:

MCl_2 is more ionic than MCl_4

Solution:

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Metal atom in the lower oxidation state forms the ionic bond and in the higher oxidation state the covalent bond because higher oxidation state means small size and great polarizing power and hence greater the covalent character. Hence MCl_2 is more ionic than MCl_4.

Q.6 Answer:

lead reacts with water containing air to form $Pb(OH)_2$

Solution:

$$2\ Pb + 2H_2O + O_2 \rightarrow 2\ Pb(OH)_2$$

Q.7 Answer:

$TlCl_5$

Solution:

Because TI^{+5} does not exist

Q.8 Answer:

14

Solution:

$SnCl_4$ is colourless volatile liquid and
$SnCl_2$ is colourless solid Sn conducts electricity and it belongs to 14 group

Q.9 Answer:

None of these

Solution:

CCl_4 is tetrahedral in nature.

Q.10 Answer:

$GeBr_4$

Solution:

The thermal stability of tetrahalides decreases in order $CX_4 > SiX_4 > GeX_4 > SnX_4$ and in terms of
same metal with different halides is in order
of$MF_4 > MCl_4 > MBr_4 > MI_4$

Q.11 Answer:

SO_2 and CO_2 are evolved

Solution:

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Charcoal is a pure form of carbon, its reaction with hot conc. H_2SO_4 is as follows:

$C + 2H_2SO_4 \longrightarrow 2H_2O^2 + 2SO_2 + CO_2$

Q.12 Answer:

it reacts with haemoglobin in RBC

Solution:

CO react with haemoglobin, forms carboxy haemoglobin and stopes the supply of O_2

Q.13 Answer:

1,3,4

Solution:

Here 1, 3 and 4 are correct as Graphite has higher electrical and thermal conductivities than diamond.

All C- atoms in graphite are sp^2 hybridised so due to partial double bond character in graphite, bond order will be more than 1 but in diamond Bond order is one.

Here only statement 2 is incorrect as diamond is more hard than graphite.

Q.14 Answer:

BN

Solution:

Graphite and boron nitride have similar structure

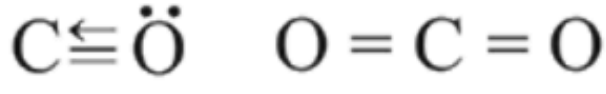
Q.15 Answer:

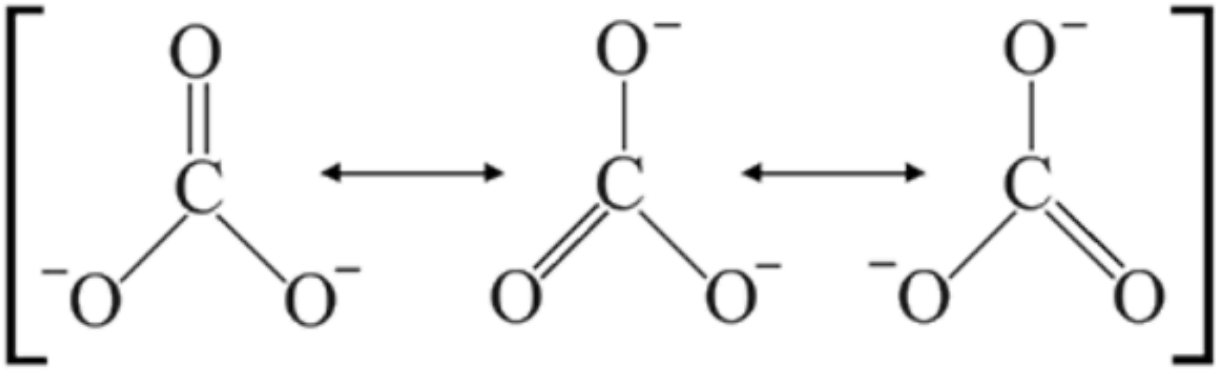
$CO < CO_2 < CO_3^{2-}$

Solution:

Structures of CO, CO_2 and CO_3^{2-} are :

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Compound	CO	CO_2	CO_3
Bond order	3	2	1.33

Bond order $\propto \frac{1}{\text{Bond length}}$ **Hence, the decreasing** $(C-O)$ **bond length is:**

$CO < CO_2 < CO_3^{2-}$

Q.16 Answer:

Only (ii)

Solution:

Carbon dioxide is not a poisonous gas.

Q.17 Answer:

C, H, S

Solution:

The elements present in organic compounds are C, H, S and their oxides are CO_2 , H_2O , SO_2 **respectively.**

CO_2 **turns lime water milky, water turns anhydrous copper sulphate blue and** SO_2 **being reducing agent can reduce** $K_2Cr_2O_7$ **into green coloured** Cr^{3+} **compound** $CrCl_3$ **etc.**

Q.18 Answer:

C^{4-} , C_2^{2-} , C_3^{4-}

Solution:

C^{4-} , C_2^{2-} , C_3^{4-} **are present in** Al_4C_3 , CaC_2 **and** Mg_2C_3 **respectively.**

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Q.19 Answer:

$348, 297, 260$

Solution:

The linking of identical atoms with each other to form long chains is called catenation. However, this property decreases from carbon to lead. Decrease of this property is associated with M-M bond energy which decreases from carbon to lead.

Q.20 Answer:

Water gas is a mixture of water vapour and hydrogen

Solution:

Water gas is $CO + H_2$

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